Introduction

Among the benchmark environments used to study RL behavior, Taxi-v3 is a classic grid-world scenario where a taxi must pick up and drop off passengers in the most efficient way. Despite its simplicity, the Taxi environment presents important RL challenges like sparse rewards, large state spaces, and the need for exploration strategies.In this project, we implemented and trained two Deep Q-Network (DQN) agents to solve the Taxi-v3 environment. The first agent uses the standard ε-greedy exploration strategy, while the second uses a softmax-based action selection method. The goal was to evaluate how exploration strategy influences learning efficiency, policy quality, and reward consistency over training.

2. Methods

2.1 Environment

We used the Taxi-v3 environment from the OpenAI Gymnasium library. It features a 5×5 grid with four fixed passenger and destination locations. The agent's task is to pick up a passenger at one location and drop them off at the correct destination while minimizing the number of steps. The state space consists of 500 discrete states, and the action space has 6 possible actions: south, north, east, west, pickup, and drop-off.

Rewards are:

- +20 for a successful drop-off,
- -10 for illegal pickup/drop-off,
- -1 for every other time step.

2.2 Exploration Strategies

To balance exploration and exploitation during training, we implemented two different strategies:

- ε-greedy strategy (used in both Q-learning and Deep Q-learning):
  The agent selects the action with the highest Q-value with probability 1 - ε, and a random action with probability ε.
  Epsilon decays exponentially using the formula:
  $\varepsilon = \varepsilon_end + (\varepsilon_start - \varepsilon_end) \cdot \exp(-decay_rate \cdot episode)$
  where:
  - $\varepsilon_start = 1.0$
  - $\varepsilon_end = 0.1$
  - $decay_rate = 0.001$
- Softmax strategy (used in tabular softmax Q-learning and softmax DQN):
  The policy uses the softmax distribution over Q-values:

$\pi(a|s) = \text{softmax}(Q(s, a) / \tau)$
where lower temperature τ sharpens preferences. We used $\tau = 1.0$ in all experiments.

2.3 Training Setup

- **Episodes:**
  - Tabular Q-learning (ε-greedy): 5000
  - Deep Q-learning (ε-greedy): 5000
  - Softmax DQN: 3000
- **Discount factor (γ):**
  - γ = **0.99** (DQN agents)
  - γ = **0.6** (tabular softmax agent)

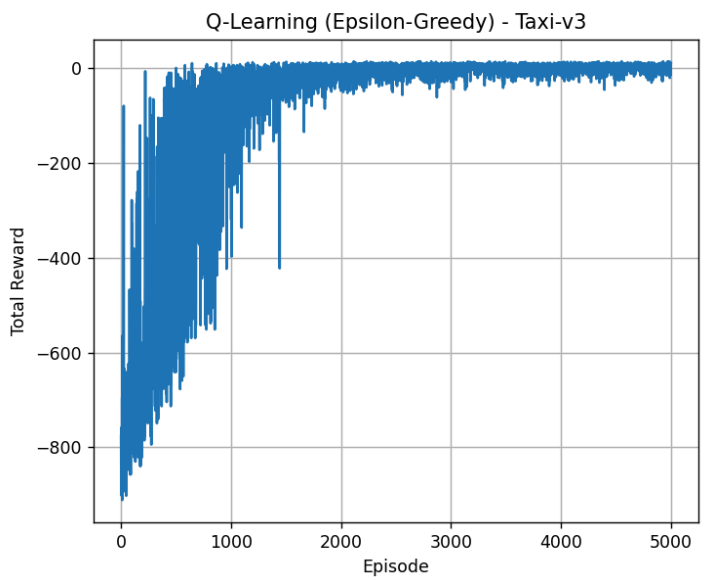
3. Results

We evaluated the Q-Learning agent using ε-greedy exploration across 5000 training episodes in the Taxi-v3 environment. The performance was analyzed using three key metrics: **episode reward**, **steps to goal**, and **epsilon decay** over time.

3.A) Q Learning (Epsilon-Greeedy)

3.1 Total Reward Over Episodes

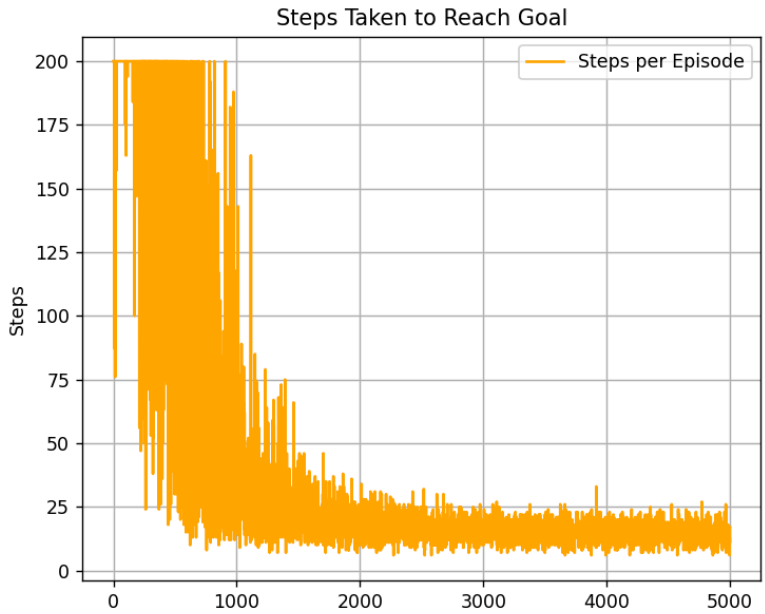
The total reward per episode increased significantly during the early episodes and gradually stabilized as learning progressed. Initially, the agent received large negative rewards due to inefficient or random actions. As the Q-values converged, the agent consistently completed tasks with improved rewards.



3.2 Steps Taken to Reach Goal

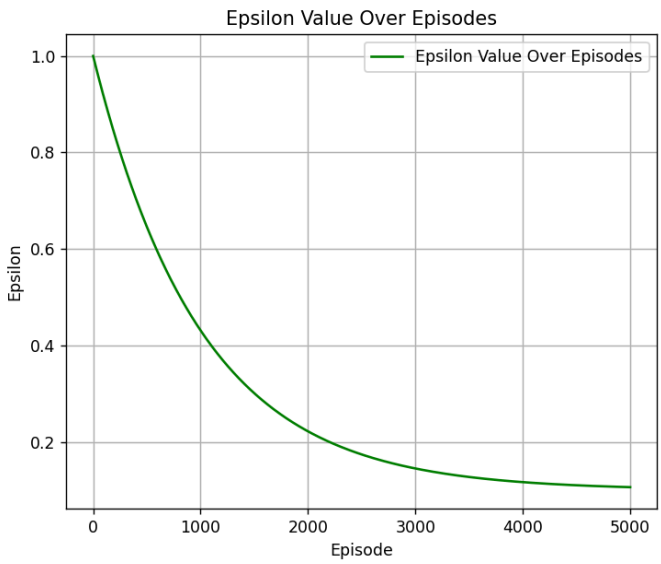
The number of steps required to reach the goal decreased dramatically throughout training. Early in training, episodes often reached the step limit (200), but by episode 1000, most tasks were completed

in under 25 steps, indicating that the agent learned an optimal or near-optimal policy.

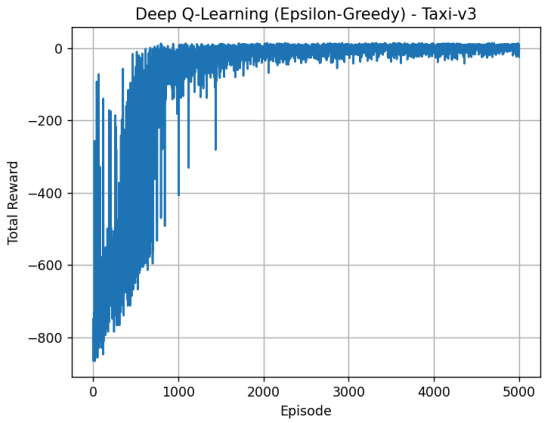


3.3 Epsilon Decay

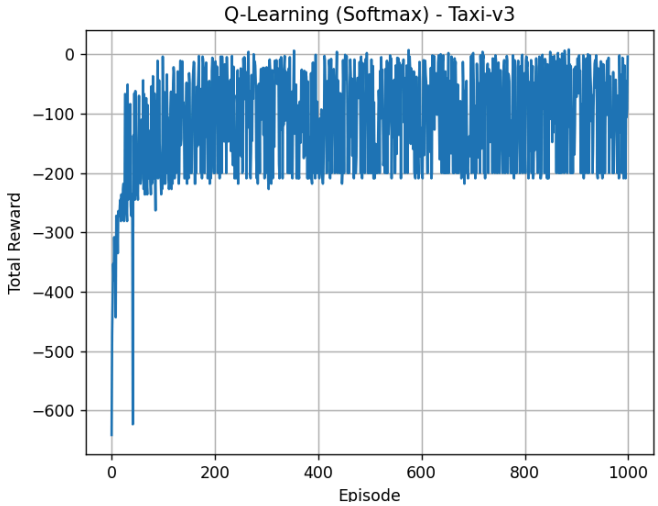
The epsilon value decayed exponentially from 1.0 to 0.1, balancing exploration and exploitation. In the initial phase, high ε promoted exploration, while later episodes emphasized exploitation of learned Q-values.



3.B) Deep Q-Learning (Epsilon-Greedy): This agent showed smooth and consistent convergence. The use of a neural network allowed for generalization over states, and the epsilon-greedy strategy efficiently balanced exploration and exploitation. The moving average reward reached approximately **+7** after 5000 episodes.



3.C) Q-Learning (Softmax): This agent used softmax-based exploration instead of ε-greedy. Its convergence was less stable, and the moving average reward plateaued at **+5 to +6**, indicating slightly lower efficiency in exploration under the chosen temperature parameter.



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